The ID3 algorithm we implemented is relatively simple, in that it has no hyperparameters (i.e., decisions about the algorithm that need to be made before running it on a given dataset). Read the scikit-learn DecisionTreeClassifier documentation and read through some of the hyperparameters that the model allows you to adjust. Pick two of the following hyperparameters:

- **min_samples_split**
- **max_features**
- **max_leaf_nodes**
- **min_impurity_decrease**

For each hyperparameter you choose, give a one-sentence explanation of what it does to affect the kinds of trees that the algorithm learns. Then, describe whether a small or large value for this hyperparameter is more likely to cause the tree to *overfit*.

An example answer to this question for the max_depth hyperparameter, (which you aren't allowed to use), might look like:

The max_depth argument controls the maximum depth (longest path from the root to any leaf) of the tree. Using a large value of max_depth is more likely to overfit, because if the tree gets so deep that each leaf only corresponds to a single training example, we wouldn't expect it to generalize well to new unseen test data.

max_leaf_nodes is a hyperparameter in decision tree algorithms that limits the maximum number of leaf nodes in a tree, affecting the kinds of trees that the algorithm learns. A small value is more likely to cause the tree to overfit as it will restrict the tree to a smaller number of leaf nodes and deeper depth.

max_features is another hyperparameter in the decision tree algorithm that limits the number of features considered for splitting at each node, affecting the kinds of trees that the algorithm learns by limiting the feature space. A small value for max_features is more likely to cause the tree to overfit leading to more complex models that are more prone to overfitting the training data.